CM52055 – Analytic software technologies

Coursework: Data Science Project

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Executive Summary

This report presents an evidence-based analysis of the National Student Survey (NSS) 2025 dataset to support strategic decision-making at the **University of Southampton** regarding the launch or enhancement of a degree program. The study analyzes student demand, satisfaction trends, and competitive edge across UK higher education institutions using exploratory and comparative analysis across 16 analytical tasks.

The analysis shows that **Business and Management**, and more specifically **Business Studies**, is the best suited subject for developing a program. **Business and Management** is the most popular subject in the country, with **51,232 students** and an average satisfaction score of **84.69%.** In this field, Business Studies is the biggest and most profitable subfield since it consistently achieved high marks for the quality of its teaching and learning materials across all the providers.

The results also show that there are clear problems with feedback systems and course organization across the board, but teaching quality is consistently and notably good. The University of Southampton already does better than the average for teaching-related NSS questions, which puts it in a good position to stand out by focusing on these weaker areas.

- The report first explains the analytical process used and evaluates how strong or robust the analysis is (Section 1).
- It then recommends a particular degree program and identifies specific areas where improvements are needed (Section 2).
- Next, it evaluates the commercial feasibility of the ideas, findings and highlights potential strategic opportunities that could be pursued (Section 3).
- Finally, it considers how the same style of analysis could be scaled up and applied to much larger National Student Survey (NSS) datasets (Section 4).

Section 1: Description of the Analysis

1.1 Dataset Overview and Context

This study uses a pre-processed subset of the National Student Survey (NSS) 2025, which collects feedback from final-year undergraduate students at publicly funded UK higher education institutions. As observed**, there are 374 providers and 158,700 records in the dataset**. It is divided into 21 broad subject areas using the Common Aggregation Hierarchy (CAH).

Each record represents a unique combination of institution, subject (across CAH1, CAH2, and CAH3 levels), and NSS survey question to make sure a uniform and comparable structure across the dataset. Student satisfaction is assessed using the **agree_pct feature,** which displays the percentage of respondents who answered "agree" or "strongly agree" for each survey question. Additional information includes the number of survey respondents, the size of the student population, and sector-wide benchmark scores.

Prior to the analysis, the dataset has been subjected to a series of validation checks to ensure essential data quality. **No missing values were observed in the dataset and each observation was confirmed to be unique at the provider-subject-question level**. Satisfaction scores and benchmark values were verified to fall within valid percentage ranges, while respondent counts were found to be consistent with reported student population. Coverage across subject areas and survey questions were comprehensive, enabling meaningful comparisons across both institutions and disciplines.

Below image shows the structure of the dataset for analyzing useful patterns and for supporting evidence -based decisions at institutional level.

```
[13]
✓ 0s
{
    "rows": nss.shape[0],
    "providers": nss["provider"].nunique(),
    "cah1_subjects": nss["cah1_subject"].nunique(),
    "cah2_subjects": nss["cah2_subject"].nunique(),
    "cah3_subjects": nss["cah3_subject"].nunique(),
    "agree_pct_present": "agree_pct" in nss.columns
}

{'rows': 158700,
 'providers': 374,
 'cah1_subjects': 21,
 'cah2_subjects': 35,
 'cah3_subjects': 160,
 'agree_pct_present': True}
```

1.2 Analytical Process and Justification

Below is a breakdown of sixteen guided tasks performed -

Phase 1: Exploratory Analysis covered in Task 1 and 2

Initial tasks focused on understanding the dataset structure, subject coverage, and provider-level distribution, using the University of Oxford as a reference case to explore total no. of subjects and subject with maximum no. of students. This stage established familiarity with the CAH hierarchy and population distributions w.r.t students.

Phase 2: Sector - Wide Satisfaction Analysis covering Task 3 to 8

This section identifies the highest and lowest performing NSS questions by average student satisfaction. It then examines top and bottom providers for these questions to understand performance extremes and competitive benchmarks. Finally, the distribution across providers were plotted for the lowest scoring questions to visualize and analyze variations between institutions.

Phase 3: Demand and Popularity Analysis covering Task 9 to 12

In this section, student population sizes were used as a proxy for demand to identify the most commercially viable subject areas. Satisfaction patterns within popular subjects were analyzed to assess the balance between demand and student feedback.

Phase 4: Strategic Subject Selection covering tasks 13 to 16 :

Finally, evidence from satisfaction, demand, and provider benchmarking was combined to recommend a specific degree program and assess the University of Southampton's competitive position. This approach hence ensured that recommendation was grounded, transparent & not biased.

1.3 Robustness and Reliability Considerations

Population figures were calculated using the sum of unique values (as per the hint in the task) to account for NSS data structure and avoid double counting across hierarchical subject classifications. While this approach does not yield exact totals but it provides a consistent measure for comparative analysis.

For distribution analysis, provider-level means were calculated first before computing overall statistics thus ensuring that results reflect institutional performance rather than being affected by subject-level repetition. Additionally, results from very small providers were interpreted carefully, as exceptionally high or low scores may not be representative of large-scale delivery program.

1.4 Main Findings from the Analysis

This analysis revealed several consistent sector-wide patterns:

- Teaching quality is the strongest-performing area across UK higher education, with **Q01 (teaching staff explaining)** showing the highest mean satisfaction (**92.85%**) and the lowest variability across providers.
- On the other hand, feedback-related questions, particularly **Q24 (feedback acted upon) and Q17 (course organization)**, record the lowest mean satisfaction scores and among the highest levels of variability across providers.
- Also low satisfaction and high variability in these areas suggests that performance is institution-specific rather than structural, indicating clear room for improvement and competitive differentiation.

- Furthermore, analysis also indicates that high student demand does not necessarily correspond to the highest satisfaction levels, presenting opportunities for institutions to outperform competitors even within popular subject areas.

Section 2: Recommending a Degree Program & Improving the NSS Scores

2.1 Recommended Degree Program

Based on the analysis of the NSS 2025 dataset, it is recommended that the **University of Southampton should launch or structurally revamp a degree program in Business Studies (CAH3 level)**.

Above recommendation is supported by clear evidence from national student demand and satisfaction patterns. At the broad subject level, **Business and Management** is the most popular subject area across UK higher education, with a total student population of **51,232 students** ranking first among all CAH1 subjects (Task 10) indicating strong and sustained demand relative to other subject areas and highlights the commercial viability of business-related program.

Within Business and Management, **Business Studies** emerges as the most appropriate CAH3-level subject for program development. It records one of the **largest student populations at CAH3 level (18,102 students)** and is offered by **127 higher education providers**, the highest provider coverage among clearly defined Business and Management sub-disciplines (Task 14). This wide adoption suggests that Business Studies represents a mature, scalable, and well-established degree model rather than a niche or specialist offering.

In terms of student experience, Business Studies achieves a **mean satisfaction score of 84.75%**, demonstrating that high demand is not associated with unusually low levels of satisfaction but also indicates a scope of improvement. While some niche subjects achieve higher satisfaction scores (for example, Welsh Studies at 96.22%, Task 12), these subjects have substantially smaller student populations and limited provider coverage. As such, they are less suitable for large-scale program development at a research-intensive university.

Overall, Business Studies offers the strongest balance between **market demand**, **institutional viability**, and **student satisfaction**, making it the most appropriate degree program for launch or enhancement at the University of Southampton.

2.2 Key Areas for Improving NSS Scores

Provider-level analysis of sector-wide NSS results highlights consistent weaknesses in specific aspects of the student experience (Task 8). The lowest-performing NSS questions at provider level are:

- **Q24 – Feedback acted upon:** 70.39% provider-level mean satisfaction
- **Q17 – Course organization:** 74.20% provider-level mean satisfaction
- **Q14 – Feedback helpfulness:** 80.98% provider-level mean satisfaction

These areas also show relatively high variability across providers, particularly for **Q24** and **Q17**, indicating that although sector-wide performance is low, significantly higher satisfaction levels are achievable in practice i.e these issues are not structural constraints of the higher education system but reflect differences in institutional practice.

Although **Q25 (students' union representation)** also appears among the lowest-performing questions , it was not prioritized for program-level intervention, as it is less directly influenced by course design and academic delivery. In contrast, feedback processes and course organization fall squarely within the scope of program management and therefore represent the most impactful opportunities for improvement.

2.3 Current Performance of the University of Southampton

The University of Southampton's NSS performance provides a strong foundation for the development of a Business Studies program. The university performs **above sector averages** in key teaching-related areas (Task 16), including:

- **Q01 – Teaching staff explaining:** 93.08%
- **Q15 – Ease of contacting teaching staff:** 89.58%

These strengths align closely with the highest-performing NSS questions within Business Studies at the sector level (Task 15), indicating that Southampton is well positioned to deliver a high-quality student learning experience in this subject area.

By contrast, questions related to **feedback responsiveness** and **course organization** do not feature among the university's strongest NSS outcomes (Task 16), mirroring the wider sector pattern. To conclude, these areas therefore represents a clear opportunity to enhance the student experience while remaining aligned with national benchmarks.

2.4 Summary of Recommendations for NSS Improvement

For a Business Studies program at the University of Southampton, the following NSS-focused priorities are recommended:

Primary improvement areas as per core importance to degree program:

- **Q24 – Feedback acted upon**
- **Q17 – Course organization**
- **Q14 – Feedback helpfulness**

Maintain existing strengths:

- **Teaching quality (Q01)**
- **Staff accessibility (Q15)**

By targeting improvements in areas where sector performance is weakest, while maintaining above-average teaching quality, the University of Southampton can improve student satisfaction in a high-demand subject area without requiring uniform improvement across all NSS dimensions.

Section 3: Commercial Feasibility & Strategic Opportunities (15 marks)

3.1.1 Market Demand for Business Studies

Analysis of student population data across UK higher education providers demonstrates strong commercial feasibility for a **Business Studies** degree program. Business and management is the **largest subject area nationally**, enrolling **51,232 students**, nearly **double** the size of the second most popular subject (social sciences: 26,742 students) and **around three times larger** than the fifth-ranked subject (law: 19,648 students).

Within the business and management category, **business studies** has the highest student population, indicating strong demand within this subject area. The scale of enrolment suggests **sustained demand and market stability**, providing a robust foundation for the launch or enhancement of a degree program in this field.

3.1.2 Provider Availability as Evidence of Market Strength

Beyond student numbers, provider coverage offers strong evidence of market validation. Business Studies is presently available at 127 higher education institutions in the UK (Task 14), indicating the broadest uptake among distinctly defined sub-disciplines of Business and Management. This wide range of offerings demonstrates that Business Studies is a developed, expandable, and firmly established degree framework, rather than a specialized or niche program.

Strong provider involvement mitigates program risk by showing consistency with industry standards, regulatory guidelines, and institutional competencies. From a business standpoint, this extensive adoption indicates that Business Studies is both practically viable and institutionally enduring across a variety of universities.

3.1.3 Competitive Position within a Sector

Despite high demand, performance within business and management varies considerably across providers. Provider-level analysis shows substantial dispersion in satisfaction scores, particularly in NSS questions where sector-wide performance is weakest.

Two areas illustrate this variability clearly:

- **Q17 (Course organization):** provider-level mean of **74.2%** with a standard deviation of **17.31%**
- **Q24 (Feedback acted upon):** provider-level mean of **70.39%** with a standard deviation of **15.61%**

The combination of low average performance and high variability indicates that while these areas represent sector-wide challenges, **high levels of performance are achievable** by some providers. This creates an opportunity for institutions to **differentiate their provision** by addressing known weaknesses in student experience.

3.1.4 Institutional Feasibility

The University of Southampton is well positioned to operate within this competitive environment. The university demonstrates a clear strength in **teaching quality**, achieving **93.08%** in **Q01 (teaching staff explaining)**, exceeding both the overall sector average (**92.85%**) and the business studies sector benchmark (**92.1%**).

This existing strength provides a strong foundation for a Business Studies program. Rather than requiring substantial improvement across all dimensions, institutional effort can be **targeted towards areas of weaker sector performance**, particularly feedback responsiveness and course organization. Focusing improvement in these areas aligns with observed student concerns while building on established strengths.

Section 4: Scaling Up for Larger Datasets

4.1 Why Scaling would be needed?

The analysis conducted in this study is based on the NSS 2025 dataset, which contains approximately **158,700 records**. While this size is manageable using in-memory data processing with pandas, the **full NSS dataset spans nearly two decades (from 2005 onwards)** and would

likely contain **tens of millions of records** when combining multiple years, institutions, subjects, and survey questions.

Scaling the analysis to this magnitude would require rethinking the analytical approach in terms of **algorithms, infrastructure, data storage, and tooling**, as outlined in the scalable data science framework introduced in the Week 11 lecture on scalable algorithms and infrastructure.

4.2 Adapting the Analysis for Larger Volumes of Data

A key principle of scalable analysis is the use of algorithms whose **time and memory requirements grow linearly or near-linearly with data size**, rather than quadratically or worse.

- Aggregations such as **group- by means, counts, sum , min , max and rankings** are naturally scalable, as they can be implemented in **O(n)** or **O(n log n)** time.
- These operations can be adapted to **streaming or batch processing**, where data is processed incrementally rather than loaded fully into memory.
- For example, the provider level means can be figured out by looking at survey responses a little at a time and updating the summary statistics. This is better than trying to store all the responses from the surveys at the same time. The provider level means are what we are trying to find so we look at the survey responses, in chunks to get the provider level means.

This approach ensures that memory usage remains bounded, even when processing datasets that exceed the capacity of a single machine.

4.3 Infrastructure and Requirements

Scaling NSS analysis to multiple decades of data would require more robust **compute, storage, and reliability infrastructure**, as described in the lecture material-

- **Compute:** Distributed CPU-based processing is sufficient for NSS-style tabular data. Frameworks such as Apache Spark or Dask allow group-by and aggregation tasks to be parallelized across multiple machines .

- **Storage:** A **data lake architecture** using columnar formats (e.g. Parquet) would allow efficient storage and retrieval of large historical datasets.
- **Reliability and scalability:** Auto-scaling clusters and fault tolerance ensure that large batch jobs can run reliably without manual intervention, even in the presence of node failures.

Such infrastructure enables efficient analysis of high-volume historical survey data without sacrificing performance or reliability.

4.4 Proposed Technology Stack

The lecture emphasizes the importance of selecting an appropriate **tech- stack**, defined as the combination of languages, frameworks, processing engines, and orchestration tools used for data analysis

A suitable scalable tech-stack for long-term NSS analysis would include:

- **Languages:** Python, to maintain continuity with the current analysis.
- **Distributed processing:** Apache Spark or Dask for scalable group-by operations and aggregations.
- **Workflow orchestration:** Tools such as Airflow or Prefect to automate ingestion, cleaning, aggregation, and reporting across multiple NSS years.
- **Data access:** SQL-based query engines (e.g. Trino or BigQuery) to support efficient filtering and summary queries across large datasets.

This stack supports reproducibility, maintainability, and efficient collaboration across analytical teams.

4.5 Tools and Workflow for Large Scale Analysis

At scale, analytical tools play a critical role in ensuring usability and reproducibility. For large-scale NSS analysis:

- **Exploration and analysis** can continue through notebook-based workflows (e.g. Jupyter), connected to distributed backends rather than local memory.
- **Version control** (e.g. Git) ensures that analytical logic remains consistent across years.
- **Dashboards and reporting tools** (e.g. Tableau or Power BI) can be used to track trends in student satisfaction over time, across providers and subjects.
- **Data quality tools** can monitor schema consistency and detect anomalies as new NSS data is added annually.

This toolset enables analysts and decision-makers to interact with large-scale results without needing to understand underlying infrastructure complexity.

4.6 Summary

In summary, scaling the NSS analysis from a single-year dataset to multiple decades of data would require:

- The use of **scalable algorithms** with linear or near-linear complexity,
- **Distributed infrastructure** capable of handling large volumes of historical data,
- A robust **tech-stack** that supports parallel processing and automation,
- And appropriate **tools** to ensure reproducibility, collaboration, and effective communication of results.

The principles and approaches outlined above closely follow the scalable data science framework presented in the Week 11 lecture and provide a realistic, well-founded strategy for extending the current analysis to large-scale NSS datasets over time .

Conclusion

This report shows how looking closely at the NSS 2025 dataset can help the University of Southampton make decisions about its programs. The University of Southampton finds that Business Studies is the choice for a degree program because it is very popular with students they are really happy with it. It fits well with what the University of Southampton is good, at.

The analysis further highlights that meaningful competitive advantage can be achieved by addressing sector-wide weaknesses in feedback and course organization. By targeting these areas while maintaining excellence in teaching quality, the University of Southampton can deliver a Business Studies program that is both commercially viable and responsive to student needs.

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